Résumé: Dr. Matthias Mittner (born Ihrke)

Contact Information

Matthias Mittner (born Ihrke)

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Phone: +47 776 46371 (office)
E-mail: `matthias.mittner@uit.no`
Web: `http://ihrke.github.io/`

Personal Information

Date of birth: September 1, 1982
Place of birth: Eberswalde, Germany
Nationality: german
Marital Status: Married to Lilli Mittner
Children: Son Kuno, Daughter Josefine

Education

Dr. phil. (summa cum laude) — 21.12.2011
University of Frankfurt, Germany
Thesis: "Binding and Retrieval of Stimulus-Response Episodes in the Negative Priming Paradigm"

B. Sc. in Computer Science — 2005/07 – 2008/04
Institute of Computer Science, University of Göttingen, Germany
Thesis: "Single Trial Estimation and Timewarped Averaging of Event-Related Potentials"

Diploma in Psychology — 2005/07 – 2007/11
Department of Psychology, University of Göttingen, Germany
Thesis: "Negative Priming and Response-Relation: Behavioural and Electroencephalographic Correlates"

Student at the University of Oslo — 2004/08 – 2005/06
Oslo, Norway
Erasmus-Scholarship; studies of computer science and psychology

B. Sc. in Psychology (equivalent) — 2002/09 – 2004/07
Department of Psychology, University of Göttingen, Germany
Thesis: "An Adaptive Approach to Memory-Span Tasks"

High-School Diploma — 2002/06
Alexander von Humboldt Gymnasium, Eberswalde, Germany
main subjects physics and mathematics

Employments

Associate Professor — 2014/04 – present
Institute of Psychology, University of Tromsø, Tromsø, Norway
Cognitive Neuroscience

Post-Doctoral Researcher — 2013/04 – 03/2014
Cognitive Science Center Amsterdam, Universiteit van Amsterdam, The Netherlands

Cognitive Neuroscience (Birte Forstmann)

Post-Doctoral Researcher 2012/09 – 2013/03
Max-Planck Institute for Experimental Medicine, Göttingen, Germany
Theoretical Neuroscience (Robert Gütig)

PhD-Student 2008/10 – 2012/01
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)
Doctoral Program: Theoretical and Computational Neuroscience (PTCN)

Guest Researcher 2008/03 – 2008/09
Max-Planck Institute for Dynamics and Self-Organization, Göttingen, Germany
cognitive modelling and data-mining (EEG)

Guest Researcher 2007/11 – 2008/02
Max-Planck Institute for Informatics, Saarbrücken, Germany
section for computer graphics; optical modelling and psychophysics

Research Assistant 2005/09 – 2007/10
Bernstein Center for Computational Neuroscience, Göttingen, Germany
cognitive modelling; psychophysical experiments

Student Assistant 2003/09 – 2004/08
Departement for Developmental Psychology, University of Göttingen, Germany
computer administration and development of experimental software

Scientific Activities

Professional Memberships
Member of the Society for Neuroscience
Program Committee Member "International Conference on Intelligent Data Engineering and Automated Learning" 2010 and 2011
Member of the Research Priority Program Brain and Cognition at the Cognitive Science Center Amsterdam (CSCA)

Organizing
Organizer for Methods-course "Introduction to Python programming" (2012)
Organizer for Methods-course "Matlab and Python programming" (2011)

Conferences
Talks and Posters at national and international conferences (CNS 06/07, BCCN symposia, TeaP, ICP2008, IDEAL, IJMC, NWG, SfN, MathPsych, ICON) – please see my website for a complete list

Advanced Courses
Course on Computational Neuroscience at the Max-Planck-Institute for Dynamics and Self-Organization 2005

DGPs Course on Recording and Analysis of Eye-Movements (Niels Galley), Cologne 2008

Reviewing
Reviewer for Spring Conference on Computer Graphics 2008 (SCCG08)
Ad-hoc Reviewer for Cognitive Neuropsycholoy, NeuroImage, Journal of Cognitive Neuroscience
IDEAL 2010/2011 (PC-member)

PUBLICATIONS

Journal Articles

Mittner, M., Boekel, W., Tucker, A. M., Turner, B.M., Heathcote, A. and Forstmann, B.U. (accepted). When the brain takes a break: A model-based analysis of mind wandering. *Journal of Neuroscience*.

Turi, Z., *Mittner, M.*, Opitz, A., Popkes, M., Paulus, W., and Antal, A. (2014). Transcranial direct current stimulation over the left prefrontal cortex increases randomness of choice in instrumental learning. *Cortex*, 63:145–154.

Mittner M., Behrendt J., Menge U., Titz C. M., Hasselhorn M. (2014). Response-Retrieval in Identity Negative Priming is Modulated by Temporal Discriminability. *Frontiers in Psychology.* 5:621.

Mittner, M. (2013). Functional Integration of Large-Scale Brain Networks. *The Journal of Neuroscience*, 33(48):18710-18711

Schrobsdorff H., *Ihrke M.*, Behrendt J., Hasselhorn M., Herrmann J.M. (2012). Inhibition in the Dynamics of Selective Attention: An Integrative Model for Negative Priming. *Frontiers in Psychology* doi:10.3389/fpsyg.2012.00491

Ihrke M., Behrendt J., Schrobsdorff H., Visser I., Hasselhorn M. (2012, accepted). Negative Priming Persists in the Absence of Response-Retrieval. *Experimental Psychology.*

Schrobsdorff H., *Ihrke M.*, Behrendt J., Herrmann J. M., Hasselhorn M. (2012). Identity Negative Priming: A Phenomenon of Perception, Recognition or Selection? *PLoS ONE 7* (3), `doi:10.1371/journal.pone.0032946`.

Ihrke, M. & Behrendt, J. (2011). Automatic generation of randomized trial sequences for priming experiments. *Frontiers in Psychology, 2* (255), `doi:10.3389/fpsyg.2011.00225`.

Ihrke, M. & Brennen, T. (2011). Sharing one biographical detail elicits priming between famous names: Empirical and computational approaches. *Frontiers in Psychology*, 2 (75), `doi:10.3389/fpsyg.2011.00075`.

Ihrke, M., Schrobsdorff, H. & Herrmann, J.M. (2011). Recurrence-Based Estimation of Time-Distortion Functions for ERP Waveform Reconstruction. *International Journal of Neural Systems*, 21 (1), pp. 65-78.

Ihrke, M., Behrendt, J., Schrobsdorff, H., Herrmann, J.M. & Hasselhorn, M. (2011). Response Retrieval and Negative Priming: Encoding and Retrieval Specific Effects. *Experimental Psychology*, 58 (2), pp. 154–161.

Behrendt, J., Gibbons, H., Schrobsdorff, H., *Ihrke, M.*, Herrmann, J.M. & Hasselhorn, M. (2010). Event-Related Brain Potential Correlates of Idendity Negative Priming From Overlapping Pictures. *Psychophysiology*, 47 (5), pp. 921–930.

Ritschel, T., Ihrke, M., Frisvad, J.R., Coppens, J., Myszkowski, K. & Seidel, H.P. (2009). Temporal Glare: Real-Time Dynamic Simulation of the Scattering in the Human Eye. *Computer Graphics Forum.* 28 (2). pp. 183-192.

Ritschel, T., Smith, K., Ihrke, M., Grosch, T., Myszkowski, K., & Seidel, H. P. (2008). 3D Unsharp Masking for Scene Coherent Enhancement. *ACM Transactions on Graphics (Proceedings SIGGRAPH 2008).* 27 (3).

Schrobsdorff, H., Ihrke, M., Kabisch, B., Behrendt, J., Hasselhorn, M., Herrmann, J. M. (2007). A Computational Approach to Negative Priming. *Connection Science*, *19* (3), 203-221.

Book Chapters

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2009). Denoising and Averaging Techniques for Electrophysiological Data. In Wennberg, R. & Perez-Velazquez, J. L. (Eds.), *Coordinated Activity in the Brain: measurements and relevance to brain function and behaviour.* Springer.

Peer-Reviewed Conference Proceedings

Schrobsdorff H., *Ihrke M.*, Herrmann J.M. (2012). Modeling Structure and Dynamics of Selective Attention. *Advances in Intelligent Systems and Computing* (Biologically Inspired Cognitive Architectures 2012), 196, pp. 287–295.

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2009). Recurrence-Based Synchronization of Single Trials for EEG-Data Analysis. *Lecture Notes in Computer Science* 5788, Intelligent Data Engineering and Automated Learning - IDEAL 2009. pp. 118-125.

Ihrke M., Ritschel, T. , Smith, K., Grosch, T., Myszkowski, K., & Seidel, H. P. (2009). A Perceptual Evaluation of 3D Unsharp Masking. *Proceedings Human Vision and Electronic Imaging XIII.*

Ihrke, M., Schrobsdorff, H. & Herrmann, J. M. (2008). Compensation for Speed-of-Processing Effects in EEG-Data Analysis. *Lecture Notes in Computer Science (Proceedings IDEAL 2008).* 5326, pp. 354-361.

Yoshida, A., Ihrke, M., Mantiuk, R. & Seidel, H. P. (2008). Brightness of the Glare-Illusion. *Proceedings of ACM Symposium on Applied Perception in Graphics and Visualization.* pp. 83-90.

Teaching

Contributing lectures and supervision (University of Tromsø) — from 2014

Undergraduate Level
– Cognitive Neuroscience (PSY-2549)
– Affective Neuroscience (PSY-2549)
– Biological Psychology (PSY-1004)
– Psychometrics (PSY-1504)
– Practical Course: Gerontopsychology (PSY-2543)

PhD Courses
– Research Design and Statistics (PSY-8002)

Introduction to Neuroscientific Methods and Brain Anatomy — summer 2013
Master-level course (University of Amsterdam)
lecture and practical on electroencephalography

Linear Regression (PSY-8012) — from spring 2015
PhD course (10 credit points)

Introduction to Python Programming (A123) — winter term, 2011/2012
GGNB methods course (introductory level), University of Göttingen

Gestaltung von Lehr-Lernprozessen — summer term, 2010
Introductory seminar to the psychology of learning and teaching for students pursuing teaching certification; University of Göttingen

Matlab and Python Programming (A109) winter term, 2010/2011
GGNB methods course (introductory level); University of Göttingen

Students

PhD students

Zsolt Turi (2011–)	"Methodological and cognitive aspects of transcranial electrical stimulation", University of Göttingen (co-supervision)
Wouter Boekel (2013–)	"When the brain takes a break: Neural correlates of mind-wandering", University of Amsterdam (co-supervision)
Leif Kåre Lende (2014–)	"Development of numerical proficiency in preschoolers", University of Tromsø(co-supervision)

Master/Diploma students

Are Svend (2014/2015)	"Title", UiT
Ida Marie Opdahl (2014/2015)	"Wandering body, wandering mind? The relationship between bodily movement, creativity and mind-wandering.", University of Tromsø
Ira Beckmann (2010)	"Negatives Priming als Identifikations- oder Selektionsphänomen", University of Göttingen

Bachelor students

Christian Fossheim (2015)	"The role of executive functions in mind-wandering"

Grants

HelseNord Post-Doc project 2015–2018
3 million NOK (≈ 350.000 EUR)
Optimizing transcranial direct current stimulation for treatment of chronic pain and depression
(3 year full-time post-doc project)

Personal Scholarships

GGNB Fellowship 2009/05–2012/01
Full–Time PhD Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

GGNB Bridging Fund Scholarship 2008/10–2008/12
Scholarship offered by Göttingen Graduate School for Neurosciences and Molecular Biosciences (GGNB)

DFG Scholarship 2009/01–2009/04
Full–Time Scholarship "Passungsverhältnisse schulischen Lernens: Verstehen und Optimieren"

Erasmus Scholarship 2004/07–2005/08
Scholarship for studies abroad, Oslo (UiO), Norway

Skills

Programming Languages
Very Good: Python, R, C, C++, Matlab/Octave
Good: Perl, Java, SQL, PHP, Presentation
Moderate: F77, Assembler, Objective C, Bash, Javascript

Computing
Parallel Programming (MPI), GPU Programming (OpenGL),
Unix/Linux Administration, LaTeX, HTML/CSS

Languages	*German* (native) *English* (fluently written, read and spoken) *Norwegian* (fluently written, read and spoken) *French* (basic skills) *Dutch* (basic skills)
Sports	running, judo, rock-climbing, skiing, hiking, table-tennis, football